The Journal of Open Source Software

Liblsl.dart: A Dart native API for Lab Streaming Layer (LSL)

Luke Daniel Ring [1], **Anna Zamm** [1], **Chris Mathys** [2], **and Simon Lind Kappel** [3]

1 School of Communication and Culture, Department of Linguistics, Cognitive Science and Semiotics, Aarhus University, Denmark **2** School of Culture and Society - Interacting Minds Centre, Aarhus University, Denmark **3** Department of Electrical and Computer Engineering - Biomedical Engineering, Aarhus University, Denmark

DOI: 10.xxxxxx/draft

Software

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Editor:

Submitted: 10 October 2025
Published: unpublished

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Summary

The `liblsl` Dart package is the first implementation of Lab Streaming Layer (LSL) in the Dart and Flutter ecosystem, enabling researchers to deploy multi-modal LSL-enabled data acquisition applications across mobile (iOS, Android) and desktop (Linux, macOS, Windows) platforms from shared source code. Unlike existing LSL implementations that are platform-specific or restricted to desktop devices (Kothe et al., 2019), this package leverages Dart/Flutter's cross-platform capabilities while maintaining the microsecond-level latency requirements of neurophysiological research through direct Foreign Function Interface (FFI) bindings to the native LSL C library (Stenner et al., 2023).

Statement of need

Neuroscience and behavioral research increasingly make use of consumer hardware due to its lower cost and sufficient performance for running experiments (Roque et al., 2025), however, the integration of consumer devices into laboratory data acquisition pipelines often requires platform-specific development and adds complexity to the data alignment and collection process, both of which can hinder flexibility and reproducibility (Iwama et al., 2024). The LSL system partially adresses some of the complexity via a software based unified method of time-synchronized data acquisition across heterogeneous hardware and software systems (Kothe et al., 2025). LSL handles clock synchronization, network communication, and data buffering, making it particularly valuable for applications requiring precise temporal alignment of multiple data sources. The `liblsl` Dart package further reduces the complexity by enabling researchers to deploy LSL-enabled applications across all major platforms from a single Dart/Flutter codebase. Potential use cases include general multimodal experiments (Dolmans et al., 2020; Wang et al., 2023), mobile brain-computer interfaces using an EEG headset and a smartphone (Blum et al., 2017; Debener et al., 2015; Stopczynski et al., 2014), hyperscanning studies (Zamm et al., 2024) that simultaneously collect neuroimaging data from multiple participants [Luft et al. (2022); Boggio et al. (2023); roqueRealTimeMobileEEG2025], and distributed experiments where multiple labs using a variety of devices collect methodologically consistent data in a standardized format for analysis (Demazure et al., 2021; Schultz et al., 2021).

Performance

Methods

Performance was characterized under controlled conditions to isolate different sources of latency. Local device tests measured the computational overhead of the package's Dart API by having a single device both produce and consume 1000 Hz data streams (16 channels, float32 format), representing typical EEG recording parameters. Network tests used two iPad Pro M4s connected via 1Gbps USB-C Ethernet adaptors to a consumer-grade gigabit router. All tests were run for 3 minutes (approximately 180,000 samples) to ensure statistical reliability. Measurements represent end-to-end latency from sample production to consumption, including API call overhead, serialization, network transmission, and deserialization.

Results and Discussion

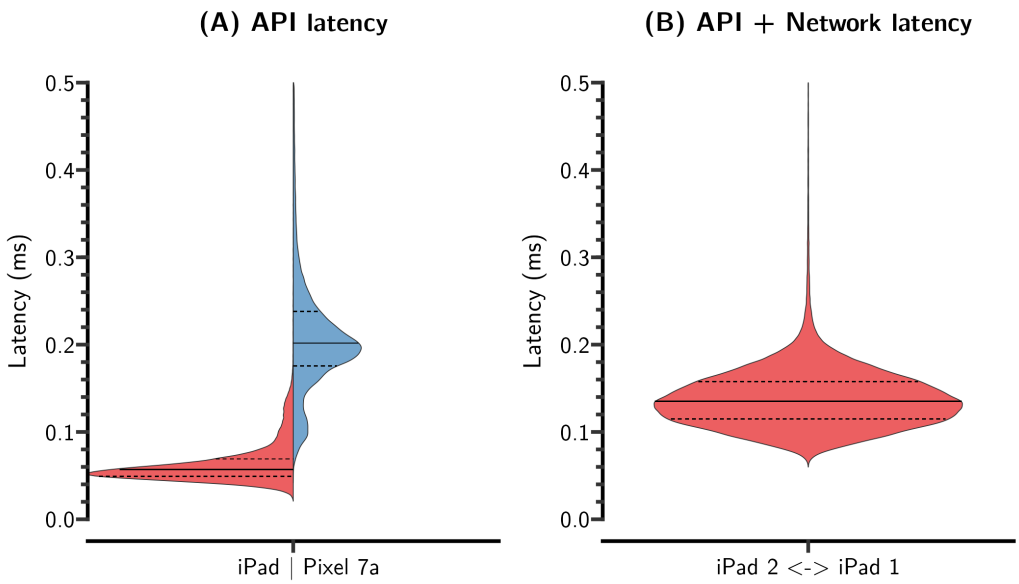

Figure 1: Latency characterization of the liblsl Dart package. **(A)** Local performance on iPad Pro M4 (left) and Pixel 7a (right), each producing and consuming its own 1000 Hz data stream. **(B)** Network performance between two iPads over 1Gbps Ethernet. Violin plots show median (solid line) with first and third quartiles (dashed lines). See Methods for experimental details and Table 1 for complete statistics. *Note: Outliers >500μs were excluded from visualization but are included in statistics.*

Table 1: Summary of latency measurements.

Condition	Device	n	Min (μs)	Max (μs)	Mean (μs)	SD (μs)
Local	iPad Pro M4	180000	21	4264	65	47
Local	Pixel 7a	179900	42	18809	274	434
Network	iPad Pro M4	180000	60	4261	148	129

Table 1 summarizes latency statistics across conditions and devices. Figure 1 illustrates the distribution of the relevant performance characteristics: (A) single-device performance showing the API's computational overhead when producing and consuming data locally (iPad: μ=65μs, σ=47μs; Pixel 7a: μ=274μs, σ=434μs), where standard deviations reflect timing jitter inherent to the operating system's thread scheduling and (B) network performance between two iPads

over a 1Gbps wired connection (μ=148μs, σ=129μs). These results confirm that the Dart wrapper introduces minimal overhead beyond the base LSL C library performance, with observed differences primarily attributable to device hardware capabilities and network infrastructure rather than the API implementation itself.

The observed latencies are consistent with previous benchmarking of native LSL implementations (Kothe et al., 2025) and demonstrate that this package preserves the real-time performance characteristics required for neurophysiological applications (Iwama et al., 2024). The local processing latencies (65-274μs) are well below typical neurophysiological event timing requirements (<1ms), confirming suitability for EEG, electromyography (EMG), and other biosignal applications[1]. The mean inter-device network latency (148μs) demonstrates that network overhead remains minimal on well-configured local networks[2], though researchers should note that wireless networks and other factors including network traffic congestion may introduce additional jitter. The device-dependent variation (iPad: 65μs vs Pixel 7a: 274μs) reflects differences in CPU architecture, device network hardware and operating system scheduling rather than API limitations, indicating that platform selection should be validated through careful testing based on the requirements of the specific application.

Example

The following code demonstrates how a complete data streaming application can be built in under 30 lines of code:

```dart
import 'package:liblsl/lsl.dart';
import 'dart:async';

void main() async {
  // Describe the stream
  final info = await LSL.createStreamInfo(
    streamName: 'Counter',
    streamType: LSLContentType.markers,
    channelCount: 1,
    sampleRate: LSL_IRREGULAR_RATE,
    channelFormat: LSLChannelFormat.int8,
    sourceId: 'uniqueStreamId123',
  );
  // Create the outlet
  final outlet = await LSL.createOutlet(streamInfo: info);

  // Send data at 2 Hz (configurable for your application)
  for (var i = 0; i < 10; i++) {
    final sample = [i];
    outlet.pushSample(sample);
    await Future.delayed(const Duration(milliseconds: 500));
  }
  // Clean up
  outlet.destroy();
  info.destroy();
}
```

[1]Raw data and source code used for generating the figures and statistics are available at: https://github.com/NexusDynamic/liblsl.dart/tree/main/packages/liblsl/paper/analysis

[2]In a lab context, this typically may be achieved by using a closed, wired network, with a high throughput hardware switch or router, and by disabling firewalls, traffic shaping and other features that may introduce latency such as deep packet inspection.

Impact

LSL lowers the complexity of multimodal time-synchronized data acquisition from multiple devices (Dolmans et al., 2020; Iwama et al., 2024; Kothe et al., 2025), and Dart/Flutter simplify cross-platform application development. Together, they provide a powerful toolset for researchers and developers to create flexible, reproducible, and accessible applications for data collection and analysis.

Acknowledgements

This library builds on liblsl by Christian A. Kothe using the Dart programming language by the Dart project authors, Google. Thanks to Chadwick Boulay and members of the dart_community Discord for help with debugging.

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